

Time-Series Momentum in a Ten-ETF Universe: Volatility Targeting, Regime-Conditioned Exposure and Crisis Performance, 2008 to 2026

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Abstract

This paper implements the time-series momentum (TSMOM) strategy of Moskowitz et al. (2012) on ten multi-asset ETFs over 222 months from April 2008 to September 2026, and decomposes it into signal, volatility targeting and a walk-forward regime overlay. Every parameter was committed to version control before any result was computed. Volatility targeting raises the net Sharpe ratio from 0.26 to 0.45 and turns monthly skewness from -0.50 to $+0.33$, but not significantly (bootstrap $p = 0.19$). The strategy gained 9.2%, 12.6% and 30.6% in the three largest equity drawdowns and tracks the AQR futures TSMOM factor ($R^2 = 0.60$). The regime overlay lowers the Sharpe ratio to 0.39 and still trails with exposure held constant; smoothed probabilities would have shown a spurious gain of 0.12. Deflated for 26 configurations, the primary Sharpe ratio is not significant. A pre-specified 20% allocation beside a 60/40 portfolio cut the maximum drawdown from 32.3% to 25.3% and the loss in all six crisis windows; its Sharpe gain, from 0.74 to 0.86, is not significant ($p = 0.08$). Trend following is supported as a crisis diversifier, not a stand-alone strategy.

1 Introduction

Time-series momentum is the tendency of an asset’s own past return to predict its future return. Moskowitz et al. (2012) (MOP) document it across 58 futures contracts from 1985 to 2009: a portfolio that is long each contract whose trailing twelve-month excess return is positive and short each contract whose return is negative earns a large and persistent premium, with its strongest returns in extreme market moves. Hurst et al. (2017) extend the evidence back over a century. Trend following of this kind is the core business of managed-futures funds, and its central selling point is “crisis alpha”: positive returns during prolonged equity sell-offs, when diversification is most valuable (Fung and Hsieh, 2001).

Two refinements are standard in practice. Volatility targeting scales positions inversely to their ex-ante volatility and the whole portfolio to a constant volatility target; Kim et al. (2016) argue that much of the TSMOM premium is in fact a volatility-scaling effect. Regime conditioning varies exposure with an estimated market state (Hamilton, 1989; Ang and Bekaert, 2002), on the premise that trend strategies earn their returns in persistent “trending” regimes and lose in choppy ones.

This paper asks four questions about a TSMOM strategy built only from liquid ETFs over 2008 to 2026, a period widely regarded as difficult for trend followers. Does the strategy earn a positive risk-adjusted return? How much of its performance comes from the signal, from volatility targeting, and from a regime overlay? Does it deliver crisis alpha in the sample’s equity drawdowns? And how robust is any of this to implementation choices, costs and multiple testing?

Every parameter (universe, sample, signal horizon, volatility estimator, risk allocation, cost assumption and regime model) was committed to version control before the first backtest was run,

so the repository history documents that nothing was tuned to the results. Alternatives appear only as labelled robustness checks, and any decision made after results were seen is identified as such.

2 Data

Universe. The universe is ten U.S.-listed ETFs in four asset classes: equities (SPY, S&P 500; EFA, developed markets excluding North America; EEM, emerging markets), U.S. Treasuries (IEF, 7 to 10 years; TLT, 20 years and over), commodities (GLD, physical gold; DBC, a diversified commodity futures basket) and currencies (UUP, long the U.S. dollar against the DXY basket; FXY, long the yen; FXA, long the Australian dollar). A euro fund was excluded as a near-duplicate of UUP (the euro is about 58% of the dollar index; daily correlation -0.95), short-duration Treasuries because their volatility of about 1.5% would need extreme leverage, and single-commodity funds as redundant with DBC and GLD.

Prices. Daily total-return prices and every distribution and split were downloaded from Yahoo Finance from 2 January 2002 to 30 September 2026. Rebuilding total returns from closing prices and distributions reproduces the adjusted series to within 6 basis points a year, the panel has no interior gaps, and every daily move larger than 8% (49 in all) was matched to an identifiable market event.

Risk-free rate and excess returns. ETFs are fully funded, whereas the futures used in the TSMOM literature earn excess returns. Every return in this paper (in signals, volatility estimates and profit and loss) is therefore an excess return over the 13-week Treasury bill rate (Yahoo Finance $\sim$ IRX), accrued daily on an actual/360 basis from the yield known at the previous close; compounded to months, it has a correlation of 0.991 with the Kenneth French one-month bill series. This matters: in 2023 and 2024 cash earned about 5% a year, and without it both the signals and the profit and loss of short positions would be misstated.

Sample. UUP begins trading in March 2007, so the first month-end at which all ten instruments have a twelve-month signal is 31 March 2008. The main sample is the 222 complete months from April 2008 to September 2026. Earlier data is used only for estimation warm-up and the regime model’s training burn-in (Section 3.5). The AQR Time Series Momentum factor, built from futures and forwards across the same four asset classes following MOP, is used as an external benchmark; it ends in May 2026, so comparisons with it cover 218 months.

Table 1 summarises the instruments. SPY has the highest Sharpe ratio (0.72). The bond funds earned little in excess of cash (1.52% a year for TLT and 1.12% for IEF), because the 2022 sell-off reversed much of the earlier bond rally, and the yen lost 3.87% a year against the dollar in excess-return terms. Appendix Table B1 reports monthly correlations. Within asset classes they are high (SPY and EFA 0.87, IEF and TLT 0.92), and the three currency funds are not independent (UUP and FXA -0.78). The effective number of independent bets within each class, measured by the eigenvalue participation ratio of its correlation matrix, is 1.26 for equities, 1.08 for rates, 1.87 for commodities and 1.93 for currencies. This motivates the risk allocation in Section 3.4.

3 Methodology

3.1 Timing convention

All signals and weights are computed from information available at the close of the last trading day of month t , and the resulting positions are held through month $t + 1$. Trades execute at that same close, as in MOP; a one-day execution lag is reported as a robustness check. Between rebalances, positions drift with prices and the portfolio is marked to market daily, so that drawdowns inside

Table 1: Instruments, April 2008 to September 2026 (222 months). Annualised mean and volatility of monthly excess returns over the T-bill rate; the Sharpe ratio is their ratio. Skewness and worst month are for monthly returns.

ETF	Class	Mean (%)	Vol (%)	Sharpe	Skew	Worst month (%)
SPY	Equity	11.18	15.61	0.72	-0.58	-16.6
EFA	Equity	5.16	17.53	0.29	-0.49	-20.9
EEM	Equity	5.07	20.94	0.24	-0.22	-25.6
IEF	Rates	1.12	6.76	0.17	0.19	-5.0
TLT	Rates	1.52	14.39	0.11	0.47	-13.1
GLD	Commodity	7.91	17.44	0.45	-0.04	-16.2
DBC	Commodity	0.90	19.11	0.05	-0.57	-25.1
UUP	FX	1.26	7.79	0.16	0.17	-6.4
FXV	FX	-3.87	9.68	-0.40	0.01	-8.5
FXA	FX	-0.57	12.40	-0.05	-0.24	-15.5

a month (most importantly March 2020) are measured.

3.2 Signals

Let $r_{i,d}$ be the daily total return of instrument i and r_d^f the bill return. The k -month signal at month-end t is the sign of the cumulative excess return over the preceding k months,

$$s_{i,t}^{(k)} = \text{sign} \left(\frac{P_{i,t}}{P_{i,t-k}} - \prod_{d \in (t-k, t]} (1 + r_d^f) \right), \quad (1)$$

where $P_{i,t}$ is the total-return price at month-end t . The twelve-month signal is the primary specification, fixed in advance from MOP. Horizons of one, three and six months and an equal-weighted blend of the four signs are reported so that the choice can be judged, not so that the best one can be chosen. Following MOP, no month is skipped.

3.3 Ex-ante volatility

Volatility is the exponentially weighted estimator of MOP,

$$\sigma_{i,t}^2 = 252 \sum_{j \geq 0} (1 - \delta) \delta^j (x_{i,t-j} - \bar{x}_{i,t})^2, \quad \frac{\delta}{1 - \delta} = 60, \quad (2)$$

where $x_{i,d}$ is the daily excess return and $\bar{x}_{i,t}$ its exponentially weighted mean. The centre of mass is 60 trading days, at least 120 observations are required, and the estimate dated t uses data up to and including the close of t . Portfolio risk uses these volatilities with the trailing 252-day correlation matrix of daily excess returns, $\Sigma_t = D_t C_t D_t$.

3.4 Strategy variants

V1: raw TSMOM. Equal notional per instrument, $w_{i,t} = s_{i,t}/N$ with $N = 10$, so gross exposure is one.

V2a: inverse-volatility weighting. Each instrument receives a share b_i of a risk budget, and its position is scaled so that its stand-alone ex-ante volatility equals that share of the 10% target σ^* :

$$w_{i,t}^{2a} = s_{i,t} b_i \frac{\sigma^*}{\sigma_{i,t}}, \quad b_i = \frac{1}{4 n_{c(i)}}, \quad (3)$$

where $n_{c(i)}$ is the number of instruments in the asset class of i . The risk budget is thus split equally across the four asset classes and then equally within each class. Because instruments within a class are highly correlated (Section 2), equal risk per instrument (the MOP convention) would give the near-duplicate equity and Treasury funds a multiple of their diversification value. This allocation was revised from equal risk per instrument before any result was computed; the per-instrument version is reported in Appendix A.

V2: volatility targeting (primary). V2a weights are rescaled so that the ex-ante volatility of the whole portfolio equals the target:

$$w_t^2 = \lambda_t w_t^{2a}, \quad \lambda_t = \frac{\sigma^*}{\sqrt{w_t^{2a'} \Sigma_t w_t^{2a}}}. \quad (4)$$

Leverage is not capped in the primary specification, mirroring the futures literature; caps are robustness checks. The budgets fix stand-alone volatility, not risk contribution, so V2 is an inverse-volatility portfolio rather than an equal-risk-contribution portfolio, and correlated same-sign positions take more than their nominal share of risk (see Section 4.3).

Benchmarks. Long-only risk parity (LO-RP) applies the V2 construction with every signal set to +1, so it isolates what TSMOM adds beyond holding the same risk-balanced, leveraged book long; Huang et al. (2020) argue that much of the TSMOM premium is of this static kind. SPY and a monthly-rebalanced 60/40 portfolio of SPY and IEF are passive benchmarks, reported without costs.

3.5 Regime-conditioned exposure

Observation series. The regime model (R1) is fitted to the weekly cross-asset trend payoff, which measures directly whether trend following is currently being rewarded. On each day d , every instrument with a twelve-month signal contributes its unit-risk TSMOM return $s_{i,m(d)} x_{i,d} / \sigma_{i,m(d)}$, where $m(d)$ is the latest month-end strictly before d (the position actually held on day d). The daily payoff is the average over the available instruments, and the weekly payoff y_w is its sum over the week.

Model. The payoff follows a two-state Gaussian Markov-switching model (Hamilton, 1989) with state-dependent mean and variance,

$$y_w = \mu_{S_w} + \sigma_{S_w} \varepsilon_w, \quad \varepsilon_w \sim \mathcal{N}(0, 1), \quad \Pr(S_w = j \mid S_{w-1} = i) = p_{ij}, \quad (5)$$

estimated by maximum likelihood with 20 random starting values, the best of which is kept.

Walk-forward estimation. At every month-end t the model is re-estimated on all weeks ending on or before t , and the regime signal is the *filtered* probability $\Pr(S_w = j \mid y_1, \dots, y_w)$ of the last such week. Filtered probabilities use only past and current observations. Smoothed probabilities, $\Pr(S_w = j \mid y_1, \dots, y_T)$, condition on the whole sample and are never used for any reported result. Because state labels are arbitrary in estimation, they are re-identified at every refit: the target (“trending”) state is the one with the higher mean payoff. At least 156 weeks are required. To have the overlay live for the 2008 crisis, the model is trained from 2003 on whichever instruments already have signals, so the number of instruments in the payoff rises from two (SPY and EFA) in 2003 to all ten by April 2008. A version trained on main-sample weeks only, and a version fitted to SPY returns instead of the trend payoff, are reported in Appendix A.

Overlay. Variant 3 scales the V2 weights by an exposure multiplier,

$$w_t^3 = (0.5 + p_t) w_t^2, \quad (6)$$

where p_t is the target-state probability at t , so exposure varies between half and one and a half

times V2. The portfolio is not re-targeted after the multiplier, which would cancel it.

Lookahead illustration. To quantify the bias that a common implementation error would create, the overlay is also computed from a single full-sample fit, using either its smoothed probabilities or its filtered probabilities. These variants use future data by construction; they are reported, clearly labelled, only as an illustration.

3.6 Backtest engine and costs

The backtest is a daily, self-financing simulation. With weights $w_{i,d-1}$ at the previous close, the portfolio's excess return on day d is

$$R_d = \sum_i w_{i,d-1} (r_{i,d} - r_d^f) - c \sum_i |w_{i,d}^{\text{target}} - w_{i,d}| \mathbb{K}_{\text{trade}} - \text{fees}_d, \quad (7)$$

and between trades the weights drift as $w_{i,d} = w_{i,d-1} (1 + r_{i,d}) / (1 + r_d^f + R_d)$. Trading costs are $c = 10$ basis points per unit of notional traded, one way, with sensitivity from 0 to 50 basis points. This is conservative for these funds, whose quoted half-spreads are typically 1 to 5 basis points, but there is no market-impact model. Borrow fees on short positions and a financing spread over the bill rate on borrowed cash are sensitivity analyses. Monthly returns compound the daily excess returns: $\prod_d (1 + r_d^f + R_d) - \prod_d (1 + r_d^f)$.

3.7 Statistical inference

Means are tested with Newey-West t -statistics using the automatic lag rule of Newey and West (1994), and Sharpe ratio standard errors follow Lo (2002). Strategy variants built from the same signals are highly correlated, so Sharpe ratio differences are tested by resampling each pair of monthly return series jointly with the stationary bootstrap of Politis and Romano (1994) (mean block six months, 5,000 replications), in the spirit of Ledoit and Wolf (2008), with a two-sided p -value. Spanning regressions measure whether a strategy earns a return beyond a static combination of benchmarks. The volatility forecast is judged by the bias statistic, the standard deviation of monthly returns divided by their ex-ante monthly volatility, which equals one for a correct forecast. The deflated Sharpe ratio of Bailey and López de Prado (2014) adjusts the primary Sharpe ratio for the number of configurations examined and for non-normal returns.

3.8 Validation of the implementation

The code base has 42 automated tests. Those most relevant to bias are truncation tests: signals, volatilities, portfolio weights, backtest returns and walk-forward regime probabilities are recomputed after deleting all data after a cut-off date (including dates in the middle of the March 2020 crash), and every value dated before the cut-off is required to be unchanged. The engine is checked against hand-calculated cases, the targeted portfolio's ex-ante volatility against an independent covariance calculation, and the gross V1 returns against the textbook formula $N^{-1} \sum_i s_{i,t-1} r_{i,t}$, which they match exactly.

4 Results

4.1 Predictability at the instrument level

Appendix Table B2 reports pooled regressions of each instrument's volatility-scaled monthly excess return, $x_{i,t} / \sigma_{i,t-1}$, on its lagged signal, with standard errors clustered by month. The coefficients rise with the horizon, from -0.001 for the one-month signal to 0.016 for the twelve-month signal, the ordering found by MOP, but none is significant at 5% (the twelve-month t -statistic is 1.66).

With instrument fixed effects every t -statistic falls, to 1.02 for the twelve-month signal, because part of the pooled relation reflects differences in average returns across instruments rather than timing, the point made by Huang et al. (2020). MOP’s lag-by-lag regressions show no more significant lags than chance would produce (Appendix Figure A1). A decomposition of the twelve-month sign strategy’s average return into a timing component and a static component (the product of average position and average return) attributes 56% to timing.

Weak instrument-level predictability is consistent with a shorter, later and narrower sample than MOP’s: 10 instruments rather than 58, over a period that includes the long, difficult stretch for trend followers in the 2010s. Whether a diversified, risk-managed portfolio earns a reliable return is a separate question, addressed next.

4.2 The three layers of performance

Table 2 reports performance net of costs, and Figure 1 the cumulative returns. The raw strategy V1 earns 1.79% a year with 7.01% volatility, a Sharpe ratio of 0.26 (Newey-West $t = 1.05$). Inverse-volatility weighting (V2a) raises the Sharpe ratio to 0.36 at much lower volatility, and targeting 10% portfolio volatility (V2) raises it to 0.45, with an annual excess return of 5.07% and a Newey-West t -statistic of 1.95. Volatility targeting also changes the shape of the return distribution: V1’s monthly skewness is -0.50 , V2’s is $+0.33$, and at a quarterly horizon V2’s skewness is $+0.82$. Positive skewness is a signature of trend following, and it appears here only after volatility scaling. V1’s negative skewness comes largely from 2009, when its equal-notional book was short equities into the March 2009 rebound, a classic momentum crash (Daniel and Moskowitz, 2016).

Table 2: Performance, April 2008 to September 2026 (222 months). Trend strategies and LO-RP are net of 10 bps costs; SPY and 60/40 are gross. Mean and Vol: annualised excess return and volatility (%). Sharpe with Lo (2002) standard error. NW t : Newey-West t -statistic of the mean. MDD: maximum drawdown of the daily cumulative excess return (%). $Skew_m$: skewness of monthly returns (normal-theory standard error 0.16). Turn.: annual turnover (multiples of NAV). Lev.: mean gross leverage at rebalance.

	Mean	Vol	Sharpe	(SE)	NW t	MDD	Calmar	$Skew_m$	Turn.	Lev.
<i>Trend strategies</i>										
V1	1.79	7.01	0.26	(0.23)	1.05	−25.8	0.07	−0.50	3.1	1.00
V2a	1.76	4.89	0.36	(0.23)	1.57	−11.7	0.15	0.17	3.3	0.90
V2	5.07	11.27	0.45	(0.23)	1.95	−26.8	0.19	0.33	9.8	2.15
V3	5.17	13.13	0.39	(0.23)	1.84	−26.3	0.20	0.40	14.2	2.55
<i>Benchmarks</i>										
LO-RP	3.19	11.00	0.29	(0.23)	1.27	−30.1	0.11	−0.35	1.9	2.07
SPY	11.18	15.61	0.72	(0.23)	3.10	−51.8	0.22	−0.58	0.0	1.00
60/40	7.15	9.65	0.74	(0.24)	3.41	−32.3	0.22	−0.62	0.2	1.00

The gains are economically meaningful but statistically fragile. Table 3 reports bootstrap tests of Sharpe ratio differences. V2 improves on V1 by 0.19 with a 95% confidence interval from -0.11 to 0.48 ($p = 0.19$). The step from V2a to V2, which adds time-varying portfolio scaling, is the closest to significance ($p = 0.06$), so the time variation in aggregate exposure, rather than the cross-sectional risk balancing alone, is where the marginal improvement sits. The choice between equal risk per asset class and per instrument makes no difference to the Sharpe ratio (Appendix Table A1). This matches the conclusion of Kim et al. (2016) that volatility scaling contributes much of TSMOM’s apparent performance, with the qualification that in this sample its contribution cannot be distinguished from noise.

Appendix Figure B1 shows the drawdowns. V2’s maximum drawdown of 26.8% ran from January 2016 to February 2019 and was not recovered until April 2022: trend following can be unrewarding for years at a time. V1’s drawdown, from March 2009, was not recovered until

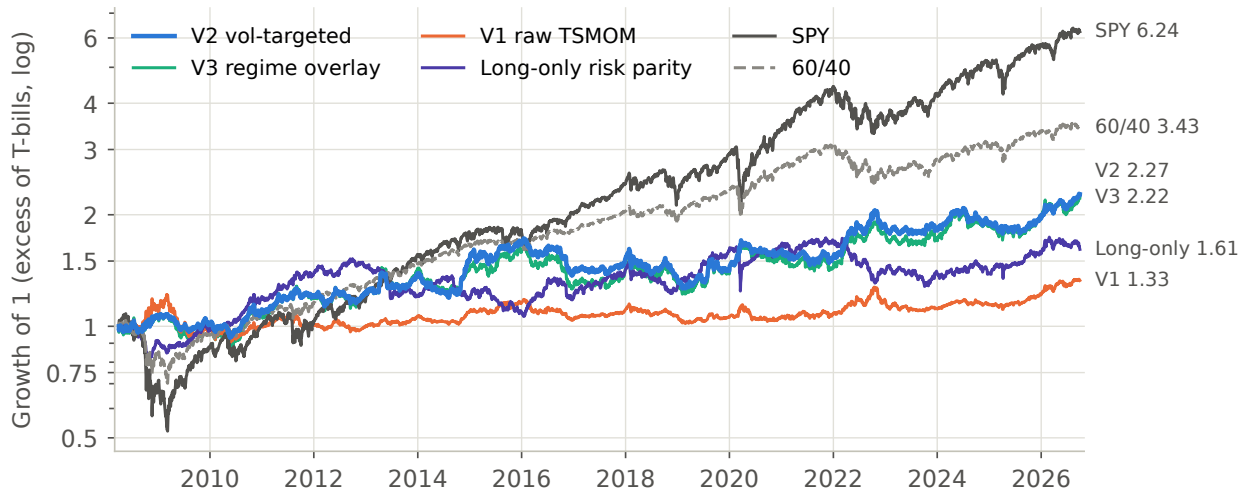


Figure 1: Growth of one unit invested in each strategy, in excess of T-bills (log scale).

Table 3: Sharpe ratio difference tests. Paired stationary bootstrap of monthly returns (mean block 6 months, 5,000 replications); two-sided p -value. Corr.: correlation of the two monthly return series. Tests for every other configuration are in Table A1.

Comparison	SR _a	SR _b	Difference	95% CI	p	Corr.
V2 vol-targeted vs V1 raw TSMOM	0.45	0.26	+0.19	[−0.11, 0.48]	0.19	0.82
V2a inverse-vol vs V1 raw TSMOM	0.36	0.26	+0.11	[−0.15, 0.35]	0.40	0.88
V2 vol-targeted vs V2a inverse-vol	0.45	0.36	+0.09	[0.00, 0.18]	0.06	0.98
V2 vol-targeted vs Long-only risk parity	0.45	0.29	+0.16	[−0.48, 0.81]	0.63	0.09
V3 regime (R1) vs V2 vol-targeted	0.39	0.45	−0.06	[−0.17, 0.05]	0.32	0.96

September 2022. Calendar-year returns show the familiar pattern of trend following: strong in 2008, 2010 and 2022, weak in 2009, 2016 and 2018.

4.3 Volatility targeting in practice

V2 targets 10% ex-ante volatility but realised 11.27%: the bias statistic is 1.13 for V2 and 1.10 for long-only risk parity, against 1.00 for V1 (Appendix Table B3). The forecasting error is not a code error: the ex-ante volatility of every rebalanced portfolio was verified independently. The overshoot comes from volatility jumps that a backward-looking estimator cannot anticipate, concentrated in September and October 2008, October 2014, February 2018 and March 2020, and because it appears in long-only risk parity too it is a property of the risk model, not the trend signal. Twelve-month rolling realised volatility ranges from about 5% to 17% (Appendix Figure B2).

The price of the target is leverage. V2’s gross leverage averages 2.15 times net asset value; its 95th percentile is 3.38, it exceeds 3 in 12% of months, and it peaks at 4.58 in the low-volatility spring of 2013. For futures this is unremarkable; for an ETF implementation it implies margin financing and short borrowing, whose costs are examined in Section 5.3. Appendix Table B4 shows how the allocation changes the source of risk. Under V1, equities supply 48.5% of ex-ante portfolio variance; under V2, each asset class supplies between 20.6% and 32.6%, with rates the largest because IEF and TLT usually hold same-sign positions.

4.4 Is it trend, static exposure, or the futures factor?

Table 4 reports spanning regressions. Regressed on SPY, V2 has a slightly negative beta (−0.10) and an annualised alpha of 6.21% ($t = 2.25$). This alpha t -statistic exceeds the t -statistic of V2’s

raw mean because the negative equity beta is credited for SPY’s strong return; it should not be read as significant stand-alone alpha. Against long-only risk parity, the more demanding benchmark, the alpha falls to 4.78% ($t = 1.76$). V2’s correlation with LO-RP is only 0.09, so the trend signal is not a disguised static risk-parity position, but the comparison is noisy and the Sharpe difference between them is insignificant (Table 3).

Table 4: Spanning regressions of monthly excess returns, $y_t = \alpha + \beta f_t + e_t$. Alpha annualised; Newey-West t -statistics. Benchmark regressions use 222 months; the AQR regressions use the 218 months to May 2026, aligned by calendar month. Correlations of V2’s asset-class sleeves with the corresponding AQR asset-class factors: rates 0.83, currencies 0.69, commodities 0.65, equities 0.63.

Strategy	Benchmark	α (%/yr)	$t(\alpha)$	β	R^2
V2	SPY	6.21	2.25	−0.10	0.02
V2	60/40	5.95	2.14	−0.12	0.01
V2	LO-RP	4.78	1.76	0.09	0.01
V1	AQR TSMOM	−0.38	−0.31	0.40	0.58
V2	AQR TSMOM	1.37	0.88	0.66	0.60

The AQR factor provides external validation. V2 has a beta of 0.66 on the AQR TSMOM factor, an R^2 of 0.60 (a correlation of 0.77) and an insignificant alpha of 1.37% a year, and each asset-class sleeve is positively correlated with the corresponding AQR sleeve. A ten-ETF portfolio therefore reproduces most of the variation of a 58-instrument futures strategy: the implementation captures the same economic phenomenon, and any shortfall in its returns reflects narrower breadth rather than a different strategy.

4.5 Crisis performance

Crisis windows are defined by a rule rather than chosen by eye: every peak-to-trough decline of more than 15% in SPY’s total-return index, measured from its running maximum within the sample. The rule yields six windows (Table 5). The 2010 decline of 15.7% is not a separate window because SPY had not yet regained its 2008 peak.

Table 5: Crisis windows: every SPY total-return drawdown deeper than 15% from its running high, April 2008 to September 2026. SPY drawdown defines the window; the remaining columns are cumulative excess returns over T-bills from the day after the peak to the trough (%). 60/40 + V2 and SPY + V2 are the 80/20 blends of Section 4.6. Lower rows: mean monthly or quarterly excess return (%) in SPY’s worst decile of months and quarters.

Peak	Trough	SPY	Excess return over the window (%)							
		drawdown	SPY	V1	V2	V3	LO-RP	60/40	60/40 + V2	SPY + V2
2008-05-19	2009-03-09	−51.5	−51.8	21.3	9.2	10.1	−16.9	−32.3	−25.3	−42.7
2011-04-29	2011-10-03	−18.6	−18.6	−6.5	−1.2	−2.3	3.5	−6.7	−5.6	−15.3
2018-09-20	2018-12-24	−19.3	−19.8	−2.5	−7.4	−7.1	−4.7	−11.1	−10.3	−17.3
2020-02-19	2020-03-23	−33.7	−33.8	6.6	12.6	9.0	−17.9	−18.9	−12.9	−25.3
2022-01-03	2022-10-12	−24.5	−25.3	16.6	30.6	29.0	−20.8	−21.6	−12.6	−15.8
2025-02-19	2025-04-08	−18.8	−19.2	−3.1	−4.1	−5.6	−8.2	−11.1	−9.7	−16.3
SPY worst-decade months, mean ($n = 23$)			−8.1	1.4	1.8	1.5	−2.6	−4.7	−3.4	−6.1
SPY worst-decade quarters, mean ($n = 8$)			−14.7	1.2	2.8	2.0	−3.2	−7.2	−5.2	−11.3

V2 was positive in the three largest drawdowns, gaining 9.2% from May 2008 to March 2009 (SPY −51.8%), 12.6% in the February to March 2020 crash (SPY −33.8%) and 30.6% over the 2022 inflation shock (SPY −25.3%). It lost in the three shorter, sharper episodes of 2011, late 2018 and early 2025, when twelve-month signals were still positioned for the preceding rally. Figure 2 shows the daily paths. In SPY’s worst decile of months, V2 averaged +1.8% against −8.1% for

SPY; long-only risk parity averaged -2.6% . Trend following tends to give back part of these gains in the recovery: V2 lost 5.7% from March to August 2020 and 8.7% between the October 2022 trough and SPY's new high in December 2023.

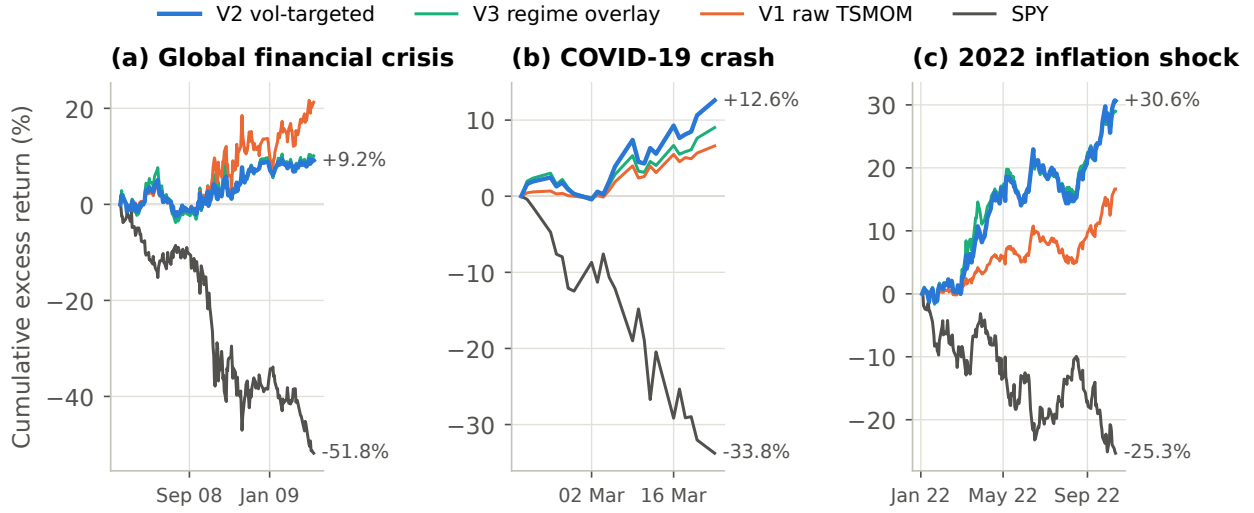


Figure 2: Cumulative excess return from the SPY peak to the trough in the three largest drawdowns.

The 2020 result contradicts a common expectation, which was also my prior: that trend followers lose in fast crashes because their twelve-month signals are still long equities. An attribution of V2's return from 20 February to 23 March 2020 shows why it did not. The equity sleeve was mixed (long SPY, short EEM, with EFA switching from long to short at the end of February) and lost only 1.2% in total, while the rest of the book was already positioned defensively: long Treasuries ($+6.6\%$), short the commodity basket ($+4.5\%$ from commodities) and long the dollar against the Australian dollar ($+2.5\%$ from currencies). Equal risk per asset class limited the damage from the one sleeve that was wrong. The same episode exposes static risk parity: on 18 March 2020 equities, Treasuries, gold and commodities fell together, and LO-RP lost 6.8% in a day.

Appendix Figure B3 plots quarterly V2 returns against quarterly SPY returns, the “smile” that Fung and Hsieh (2001) associate with trend followers, whose payoff resembles a long straddle. The fitted quadratic coefficient is positive (0.67) but not significant ($t = 1.20$, 74 quarters), so the convexity is visible but statistically weak in this sample.

4.6 Trend following as a portfolio diversifier

V2's Sharpe ratio of 0.45 is below those of SPY (0.72) and the 60/40 portfolio (0.74), so the case for holding it rests on its role as a diversifier. This subsection tests that role directly. It was added after the results above had been seen, and its specification was committed before it was computed. It uses V2 exactly as reported, so it does not add a trend configuration to the trial count of Section 5.4.

Design. The primary portfolio holds 80% in the 60/40 benchmark and 20% in V2; a secondary portfolio holds 80% in SPY and 20% in V2. Allocations of 10% to 20% to managed futures are common, and 20% was fixed in advance; no other weight was computed. The sleeves are reset to their target weights at each month-end and drift within the month, so daily drawdowns are measured as in Table 2. V2 is net of costs while the benchmarks are gross; the blend's own rebalancing (0.13 of net asset value a year) is not charged. The specification names drawdowns and crisis losses as the headline, because that is the purpose of a diversifier.

Results. V2's monthly correlation with the 60/40 portfolio is -0.11 . Adding it at 20% lowers the 60/40 portfolio's volatility from 9.65% to 7.81% and its mean less, from 7.15% to 6.74% , so the

Table 6: Trend following held beside conventional portfolios, April 2008 to September 2026 (222 months). Blends hold 80% in the benchmark and 20% in V2, rebalanced monthly with daily drift. Mean and volatility are annualised excess returns over T-bills (%); the maximum drawdown (MDD, %) and its recovery date are from daily data. Corr. is the monthly correlation of V2 with the benchmark. Δ SR is the Sharpe ratio of the blend minus that of its benchmark, with a two-sided paired stationary bootstrap p -value. Turn. is the net asset value moved between sleeves per year. V2 is net of 10 basis points; the benchmarks are gross of costs.

	Mean	Vol	Sharpe	(SE)	Max. drawdown		Calmar	Skew _m	Corr.	vs benchmark		
					MDD	Recovery				ΔSR	p	Turn.
<i>Primary: 60/40 investor</i>												
60/40	7.15	9.65	0.74	(0.24)	−32.3	2010-10-05	0.22	−0.62	−0.11			
80/20 blend	6.74	7.81	0.86	(0.24)	−25.3	2010-09-20	0.27	−0.47	−	+0.12	0.08	0.13
<i>Secondary: equity-only investor</i>												
SPY	11.18	15.61	0.72	(0.23)	−51.8	2011-04-27	0.22	−0.58	−0.14			
80/20 blend	9.95	12.37	0.80	(0.24)	−42.7	2011-01-27	0.23	−0.50	−	+0.09	0.04	0.17
All four maximum drawdowns: peak 2008-05-19, trough 2009-03-09.												

Sharpe ratio rises from 0.74 to 0.86. The maximum drawdown falls from 32.3% to 25.3%. For the equity-only investor it falls from 51.8% to 42.7%, and the Sharpe ratio rises from 0.72 to 0.80. In each of the six crisis windows, for both investors, the blend lost less than its benchmark (Table 5). The largest reductions came when V2 itself made money: the 60/40 investor's loss fell from 18.9% to 12.9% in 2020 and from 21.6% to 12.6% in 2022. The reductions in 2011, 2018 and 2025 were small (0.8 to 1.4 percentage points) and come mainly from holding 20% less of the benchmark. Figure 3 shows the full paths.

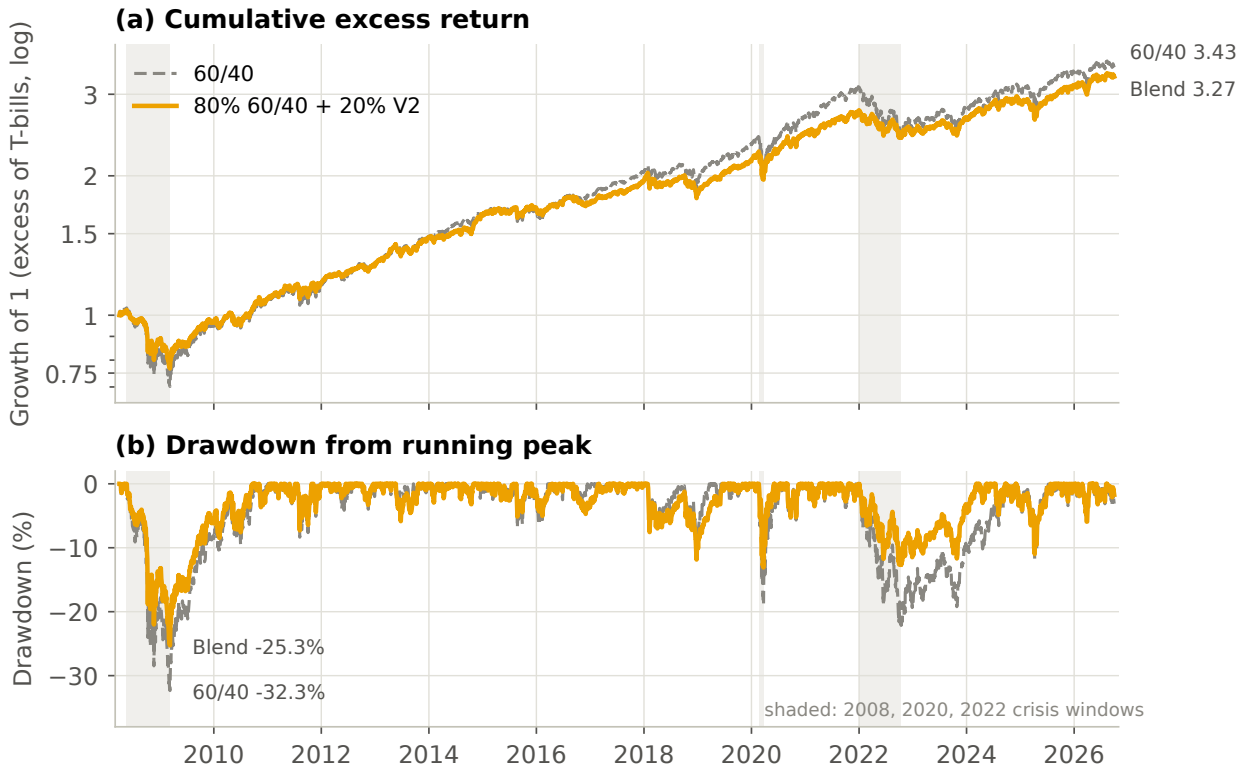


Figure 3: The 60/40 portfolio and the blend of 80% 60/40 and 20% V2, daily: (a) growth of one unit in excess of T-bills (log scale); (b) drawdown from the running peak. Shaded: the 2008, 2020 and 2022 crisis windows.

Statistical significance. The improvement in the Sharpe ratio is not significant for the

primary blend: the difference is 0.12 with a bootstrap p -value of 0.08. For the equity-only blend the difference of 0.09 has $p = 0.04$, but two blends were tested, and a Bonferroni adjustment for the pair would put it at 0.08. The evidence therefore supports a narrower claim than “trend following improves a portfolio”: over this sample, a fixed 20% allocation to V2 reduced drawdowns and crisis losses in every episode, at the cost of a lower mean return, and the rise in the risk-adjusted return is consistent with that benefit but not statistically established.

4.7 The regime overlay

Table 7 evaluates the overlay. V3 lowers the Sharpe ratio from 0.45 to 0.39 ($p = 0.32$). Regressed on V2, it has a beta of 1.12 and a negative, insignificant alpha: the overlay mainly raises average exposure (its mean multiplier is 1.19) without improving its timing.

Holding exposure constant. The comparison with V2 therefore mixes the level of exposure with its timing. Two checks, added after these results and specified before they were computed, separate the two. The first divides the multiplier by its own walk-forward average, $m_t = (0.5 + p_t)/(0.5 + \bar{p}_t)$, where $\bar{p}_t$ is the mean target-state probability from the first refit up to month t , so no future data enter (dividing by the full-sample mean would be a lookahead error). Its Sharpe ratio is 0.40 ($p = 0.33$ against V2), its alpha on V2 is negative, and m_t does not forecast V2’s returns (Table 8). The second scales V2 by the constant 1.19, a non-tradable diagnostic with V2’s Sharpe ratio by construction; V3 falls 0.06 short of it ($p = 0.32$). Under the rule fixed in advance, the overlay adds no timing value. Its lower drawdown reflects lower average exposure (mean multiplier 0.93), not better timing.

Table 7: Regime-conditioned exposure. Δ SR and p : Sharpe difference from V2 and bootstrap p -value. α on V2: annualised alpha (%) from regressing the overlay on V2, Newey-West t . V3 normalised divides the V3 multiplier by its walk-forward mean. [‡]Scales V2 by V3’s realised mean multiplier, which uses full-sample information: a diagnostic, not a tradable strategy, with V2’s Sharpe ratio by construction, so no test is reported. [†]Lookahead illustrations use a single full-sample fit and therefore future data; they are not results. Other overlay variants are in Table A2.

	Mean	Vol	Sharpe	MDD	Lev.	Δ SR	p	α on V2	$t(\alpha)$
V2 vol-targeted	5.07	11.27	0.45	−26.8	2.15	—	—	—	—
V3 regime (R1)	5.17	13.13	0.39	−26.3	2.55	−0.06	0.32	−0.52	−0.74
<i>Exposure held constant (added after the results; specified in advance)</i>									
V3 normalised (walk-forward)	4.06	10.23	0.40	−21.5	2.01	−0.05	0.33	−0.39	−0.74
V2 $\times$ 1.19 [‡]	6.04	13.43	0.45	−31.4	2.56	0.00	—	—	—
<i>Lookahead illustrations (use future data; not results)</i>									
Smoothed probabilities [†]	6.31	10.99	0.57	−18.7	2.28	+0.12	0.17	—	—
Full-sample parameters [†]	4.91	11.51	0.43	−20.1	2.24	−0.02	0.73	—	—

Two diagnostics explain the failure. First, in every non-degenerate refit the higher-mean “trending” state is also the lower-variance state, so the model acts largely as a volatility filter on the trend payoff. Second, the label does not forecast returns: the slope of V2’s next-month return on the probability is negative ($t = -0.74$), and V2’s Sharpe ratio is 0.33 in “trending” months against 0.80 in the others (Table 8, Figure 4).

Degenerate fits. In 25 of the first 27 monthly refits (March 2008 to June 2010), the maximum-likelihood estimate devotes one state to a handful of isolated extreme weeks from the burn-in period: that state has a mean payoff of about -0.27 per week, against a full-sample weekly standard deviation of 0.07, and a probability of persisting of zero. This is a known weakness of maximum likelihood for mixtures of normal distributions, whose likelihood rewards fitting individual extreme observations (Kiefer, 1978; Hamilton, 1991). The remaining state was then labelled “trending” with probability close to one, so V3 held 1.5 times V2’s exposure through 2008 to 2010; no degenerate

Table 8: Does the regime probability forecast trend returns? V2's return in month $t + 1$ regressed on the target-state probability p_t (slope in % per month per unit of probability; Newey-West t), and V2's annualised Sharpe ratio in months with p_t above and below 0.5. [‡]Regressor is the normalised multiplier m_t (slope per unit of multiplier), split at $m_t > 1$. [†]Uses future data.

Model	Slope	t	$p_t > 0.5$		$p_t \leq 0.5$	
			Sharpe	Months	Sharpe	Months
R1 primary	-0.72	-0.74	0.33	176	0.80	46
R1 normalised [‡]	-0.93	-0.71	0.31	101	0.55	121
R1 smoothed [†]	0.91	1.30	0.83	117	0.13	105

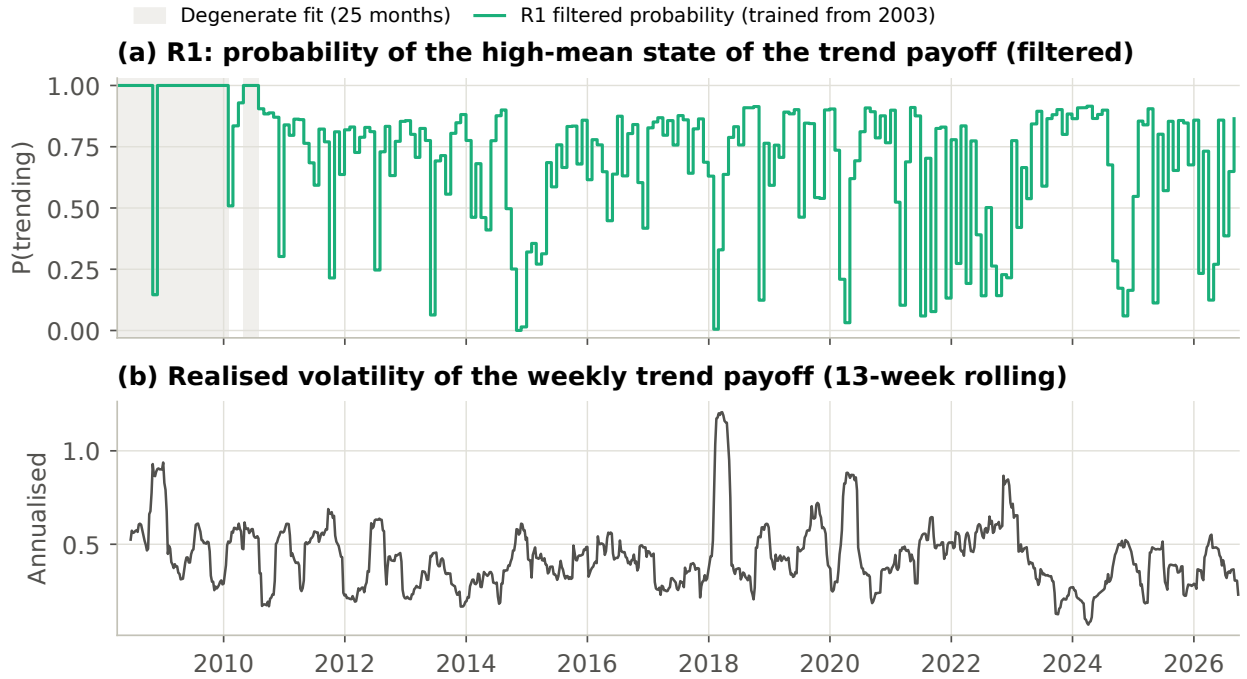


Figure 4: (a) Walk-forward filtered probability of the trending state, set at each month-end and held over the following month, with the holding months of the 25 degenerate fits shaded. (b) Realised volatility of the weekly trend payoff.

fit occurs after June 2010. The pre-registered result is reported unchanged. A diagnostic variant, defined after the problem was found, sets the multiplier to one at degenerate refits and still trails V2 (Sharpe 0.42; Appendix A).

The cost of lookahead. The same overlay computed from full-sample smoothed probabilities would have reported a Sharpe ratio of 0.57, a gain of 0.12 over V2, a maximum drawdown of 18.7% instead of 26.8%, and a positive predictive slope. Full-sample parameters with filtered probabilities produce no such gain (0.43), so the spurious improvement comes from smoothing, which conditions each week’s state on future observations, not from the parameter estimates. The gap of 0.18 between the smoothed and walk-forward versions measures how large this common error can be.

5 Robustness and Validation

5.1 Signal horizon and robustness grid

V2’s Sharpe ratio rises with the signal horizon, as in the literature, from -0.20 at one month to 0.45 at twelve months, with 0.43 for the blend of all four (Appendix Figure B4). The 95% intervals overlap heavily, and the twelve-month primary was fixed in advance rather than selected.

Table 9 varies one element of V2 at a time. Every variant is reported and none was selected; the rows not shown here are in Appendix Table A1. A one-day execution lag leaves the Sharpe ratio unchanged (0.45), so the results do not depend on trading at the signal close. Capping gross leverage at 2 times reduces the Sharpe ratio to 0.41 and volatility to 9.4% . A faster volatility estimator (centre of mass 20 days) lowers the Sharpe ratio to 0.40 ($p = 0.08$) and widens rather than narrows the volatility overshoot (bias statistic 1.19), so the overshoot is not a matter of a slow estimator. A continuous signal, $\text{clip}(t_{i,t}/2, -1, 1)$ where $t_{i,t}$ is the t -statistic of the mean daily excess return over the past twelve months (Baltas and Kosowski, 2013), has the highest point estimate in the grid (0.57), but the difference from V2 is insignificant ($p = 0.22$). It was specified during the robustness analysis, before its result was computed, and it was not adopted.

Table 9: Robustness grid: V2 with one element changed. ΔSR and p : Sharpe difference from V2 and bootstrap p -value. Corr.: correlation with V2. Vol and MDD in %. Vol bias: bias statistic of the variant’s own ex-ante volatility forecast. All net of 10 bps.

Variant	Sharpe	ΔSR	p	Corr.	Vol	MDD	Turn.	Vol bias
Primary (V2)	0.45	–	–	1.00	11.3	–26.8	9.8	1.13
<i>Implementation and estimation</i>								
Execution lag 1 day	0.45	+0.00	0.92	0.99	11.1	–30.5	9.8	1.11
EWMA centre of mass 20 days	0.40	–0.05	0.08	0.99	11.9	–27.3	11.7	1.19
EWMA centre of mass 120 days	0.46	+0.01	0.48	1.00	11.0	–27.9	9.0	1.10
Gross leverage cap $2\times$	0.41	–0.04	0.29	0.98	9.4	–22.4	6.6	1.13
<i>Signal</i>								
Continuous t -statistic signal	0.57	+0.12	0.22	0.94	11.6	–24.3	8.4	1.16
<i>Leave one asset class out</i>								
Excluding equity	0.29	–0.16	0.09	0.93	11.5	–35.0	9.9	1.15
Excluding rates	0.41	–0.04	0.77	0.85	10.5	–26.6	9.6	1.05
Excluding commodities	0.34	–0.11	0.25	0.92	10.9	–21.6	9.3	1.09
Excluding FX	0.50	+0.05	0.52	0.94	10.6	–28.8	7.3	1.06

Leaving out one asset class at a time shows where the returns come from. Removing equities lowers the Sharpe ratio most (to 0.29 , $p = 0.09$), followed by commodities (0.34). Removing currencies raises it to 0.50 , consistent with the weak trends of the currency sleeve, although the difference is again insignificant. No single asset class drives the result, but equity and commodity

trends contributed most.

5.2 Subperiods

Appendix Table B5 splits the sample into two halves of 111 months. V2’s Sharpe ratio is 0.38 in the first half and 0.52 in the second, positive in both and well within one standard error (0.33) of each other. The regime overlay is unstable across halves: V3 is weaker than V2 in the first half, when the degenerate fits occurred (0.27 against 0.38), and slightly stronger in the second (0.53 against 0.52). SPY’s Sharpe ratio exceeds V2’s in both halves.

5.3 Costs, borrowing and financing

Appendix Table B6 (and Appendix Figure B5) shows the effect of trading costs. V2 trades 9.8 times its net asset value a year, so each basis point of one-way cost costs about 0.1% a year. Its Sharpe ratio is 0.54 before costs, 0.45 at the assumed 10 basis points and 0.10 at 50 basis points; its break-even cost is 62 basis points per unit traded, well above the cost of trading these funds. The regime overlay raises turnover to 14.2 and lowers the break-even to 47 basis points. Long-only risk parity, which trades little, overtakes V2 at costs above about 30 basis points.

Financing assumptions matter more than trading costs for an ETF implementation. V2 holds short positions averaging 0.98 times net asset value, and a borrow fee of 100 basis points a year lowers its Sharpe ratio from 0.45 to 0.36. A financing spread of 100 basis points over the bill rate on borrowed cash, which averages 0.33 times net asset value, lowers it to 0.42, and with both at 100 basis points the Sharpe ratio is 0.33.

5.4 Multiple testing

Twenty-five distinct TSMOM configurations were examined in the original study: V1 at four horizons, V2a, four versions of the regime overlay and the sixteen rows of the robustness grid (the lookahead illustrations are excluded). The cross-sectional standard deviation of their Sharpe ratios is 0.17. Against this background, the probability that V2’s true Sharpe ratio exceeds zero, adjusted for non-normality but not for selection, is 0.975. The deflated Sharpe ratio, which asks instead whether V2 beats the best Sharpe ratio expected from N independent trials with no true skill, is 0.68 for $N = 25$ (Table 10). Because the configurations are highly correlated, the effective number of independent trials is well below 25, but even with $N = 2$ the deflated Sharpe ratio is 0.94, below the conventional 0.95 threshold, and adding the later exposure-normalised overlay (Section 4.7) as a twenty-sixth configuration leaves it at 0.68. Every configuration in the trial count is listed, with its Sharpe ratio and its test against V2, in Appendix Table A1. The primary result is therefore suggestive rather than statistically established, a conclusion consistent with the cautions of Harvey et al. (2016) on backtested performance.

Table 10: Deflated Sharpe ratio of V2 (Bailey and López de Prado, 2014). SR_0 is the expected maximum annual Sharpe ratio of N independent zero-skill trials, given the observed cross-trial dispersion (0.17); DSR is the probability that V2’s true Sharpe ratio exceeds SR_0 , adjusted for the skewness and kurtosis of its monthly returns. Rows up to $N = 25$ use the dispersion of the original 25 configurations; the $N = 26$ row adds the exposure-normalised overlay to the trial set.

Number of trials N	SR_0 (annual)	DSR
2	0.09	0.94
5	0.21	0.86
10	0.27	0.78
20	0.33	0.70
25 (original study)	0.34	0.68
26 (including the normalised overlay)	0.34	0.68

6 Limitations

ETFs are not futures. A futures implementation would hold margin rather than full notional, finance leverage at close to the bill rate and short without a borrow fee. V2’s average gross leverage of 2.15 and short exposure of about one times net asset value instead need a margin account and a borrow facility, whose costs can remove a quarter of the Sharpe ratio. Margin calls, recall of borrowed shares and retail leverage limits are not modelled.

Breadth. Ten instruments, with well under ten independent bets, are far fewer than MOP’s 58. Low breadth makes returns more concentrated and the statistics noisier, and explains at least part of the shortfall against the AQR factor.

One sample, one bond cycle. The sample contains one long decline in interest rates and one sharp rise, and Treasury trends helped in both; a different rate path could give materially different results. With a Sharpe ratio standard error of about 0.23, even a true Sharpe ratio of 0.5 would be hard to distinguish from zero.

Volatility forecast. Realised volatility overshoot the target in volatility jumps, so a risk limit would have been breached in the episodes that matter most, and trailing correlations cannot anticipate a sudden break such as March 2020.

Regime model. The model has limited power with weekly data over about 20 years. Alternatives (fat-tailed errors, a minimum regime duration, more states) were not explored, because choosing among them after seeing results would be a specification search; the negative result applies to this specification only.

Multiple testing and post-results choices. The twelve-month horizon and 60-day estimator were chosen with knowledge of the literature’s results on longer samples. Four elements were added after results were seen: the degenerate-fit diagnostic, the continuous signal, the portfolio blends and the exposure-normalised overlay (the last two committed before they were computed). None changes a primary result, and the deflated Sharpe ratio counts every trend configuration.

Portfolio application. The 20% weight is fixed, not optimised. The benchmarks are gross of costs, the blend’s rebalancing is not charged, and the benefit rests on the same rate path and crises behind V2’s crisis gains.

Fund mechanics, execution and data. DBC rolls by an “optimum yield” rule and charges about 0.85% a year, costs are a flat 10 basis points with no market impact, and prices come from a free source for funds that still exist in 2026.

7 Conclusion

A time-series momentum strategy built from ten liquid ETFs earned a positive return over 2008 to 2026, and volatility targeting was the layer that made it useful: it raised the Sharpe ratio from 0.26 to 0.45, turned negative skewness positive and balanced risk across asset classes. The strategy gained in each of the three largest equity drawdowns, tracks the AQR futures factor closely, and held at a pre-specified 20% beside a 60/40 portfolio it reduced the maximum drawdown from 32.3% to 25.3% and the loss in every crisis window. Neither the gain from volatility targeting, nor the strategy’s deflated Sharpe ratio, nor the blend’s Sharpe improvement is statistically significant.

The regime overlay did not help: it lowered the Sharpe ratio, it trailed V2 even at V2’s average exposure, and its “trending” state was mainly a low-volatility state with no forecasting power. The same overlay with smoothed probabilities would have appeared to add 0.12 to the Sharpe ratio, which shows how much of the apparent value of regime models can be an artefact of using future information. The honest summary is that trend following is supported as a crisis diversifier with modest and statistically fragile stand-alone returns; a broader futures universe is the natural next step.

Reproducibility

All code is in the project repository. After the data downloads (`fetch_prices.py`, `fetch_rates.py`, `fetch_benchmarks.py`), `run_all.py` rebuilds every result in order: `check_data.py`, `run_stage2.py` (signals, volatility and predictive regressions), `run_stage3.py` (engine and V1), `run_stage4.py` (volatility targeting and benchmarks), `run_stage5.py` (regime overlay), `run_stage6.py` (evaluation, crisis, costs, robustness and deflated Sharpe ratio), `run_overlay_norm.py` (regime overlay with exposure held constant), `run_blend.py` (portfolio blends), `make_tables.py` and `make_figures.py`. Every parameter, including the blend and normalised-overlay specifications, is in `config.py`. Every table and figure is generated from the saved outputs, so no tabulated number is transcribed by hand.

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A Configurations examined but not shown in the main text

Table A1 lists every configuration that counts towards the trial number of the deflated Sharpe ratio (Section 5.4): the 25 of the original study and the exposure-normalised overlay added later. Each Sharpe ratio is the one used in the deflated Sharpe calculation. Configurations moved out of the main text to keep it short were still examined, and all of them remain in the count.

Regime overlay variants. Three overlay variants were evaluated alongside the primary V3 (Table A2). R2 fits the same two-state model to the weekly excess return of SPY, the textbook market-regime indicator, and takes the higher-variance (“turbulent”) state as the target, which tests the crisis-alpha thesis directly. The main-sample version fits R1 to main-sample weeks only, so it is live from March 2011 and has no degenerate fits. The third version, defined after the degenerate fits were found, flags a refit as degenerate when either state’s probability of persisting is below 0.5 and sets the multiplier to one at flagged refits. All three trail V2. Under the SPY model, V2 earns the same Sharpe ratio (0.45) in turbulent and calm months, so there is no evidence in this sample that trend returns are concentrated in high-volatility equity regimes. The main-sample model’s predictive slope is zero, and excluding the degenerate fits leaves the slope negative, because the clean fits do not forecast trend returns either.

Lag-by-lag predictability. Figure A1 repeats MOP’s lag-by-lag regressions of the volatility-scaled return on its own lagged value and sign. Two of the 96 regressions (both at lag 8) exceed $|t| = 1.96$, fewer than the five that chance alone would produce.

B Supplementary tables and figures

Floats referred to in the main text, in the order in which they are cited.

Table A1: Every configuration in the trial count ($N = 26$), net of 10 bps. Δ SR: Sharpe ratio minus V2’s. p : two-sided paired stationary bootstrap p -value against V2, as in Table 3; V1 at one, three and six months was not tested against V2 in the original study, so only the difference is shown. Also shown in: where else in the paper the configuration appears.

Configuration	Sharpe	Δ SR vs V2	p	Also shown in
<i>Signal and construction</i>				
V1, 1-month signal	−0.12	−0.57	–	–
V1, 3-month signal	0.26	−0.19	–	–
V1, 6-month signal	0.20	−0.25	–	–
V1, 12-month signal	0.26	−0.19	0.19	Table 2
V2a inverse-vol	0.36	−0.09	0.06	Table 2
V2 vol-targeted (primary)	0.45	–	–	Table 2
<i>Regime overlay</i>				
V3 regime (R1, primary)	0.39	−0.06	0.32	Table 2
V3-R2 regime (SPY)	0.32	−0.13	0.21	Table A2
V3 main-sample HMM	0.43	−0.02	0.64	Table A2
V3 excl. degenerate fits	0.42	−0.03	0.62	Table A2
V3 normalised (walk-forward)	0.40	−0.05	0.33	Table 7
<i>Robustness grid: V2 with one element changed</i>				
Execution lag 1 day	0.45	+0.00	0.92	Table 9
EWMA centre of mass 20 days	0.40	−0.05	0.08	Table 9
EWMA centre of mass 120 days	0.46	+0.01	0.48	Table 9
Equal risk per instrument	0.45	+0.00	0.99	–
Gross leverage cap 2×	0.41	−0.04	0.29	Table 9
Gross leverage cap 3×	0.43	−0.02	0.17	–
Continuous t -statistic signal	0.57	+0.12	0.22	Table 9
Lookback 1 month	−0.20	−0.64	0.02	Figure B4
Lookback 3 months	0.32	−0.13	0.53	Figure B4
Lookback 6 months	0.35	−0.10	0.61	Figure B4
Blend of 1/3/6/12 months	0.43	−0.02	0.88	Figure B4
Excluding equity	0.29	−0.16	0.09	Table 9
Excluding rates	0.41	−0.04	0.77	Table 9
Excluding commodities	0.34	−0.11	0.25	Table 9
Excluding FX	0.50	+0.05	0.52	Table 9

Table A2: Regime overlay variants not shown in the main text. Columns as in Tables 7 and 8: annualised mean, volatility and MDD in %, alpha from regressing the overlay on V2 (Newey-West t), slope of V2’s next-month return on the target-state probability, and V2’s Sharpe ratio in months with the probability above and below 0.5.

	Mean	Vol	Sharpe	MDD	Lev.	α on V2		Predictive slope		V2 Sharpe	
						%/yr	t	Slope	t	$p_t > 0.5$	$p_t \leq 0.5$
V3-R2 regime (SPY)	3.16	9.76	0.32	−31.7	1.52	−0.79	−0.81	−0.34	−0.53	0.45	0.45
V3 main-sample HMM	5.33	12.54	0.43	−26.1	2.45	−0.09	−0.15	0.00	0.00	0.33	0.64
V3 excl. degenerate fits	5.41	12.78	0.42	−26.3	2.48	−0.14	−0.22	−0.31	−0.28	0.42	0.74

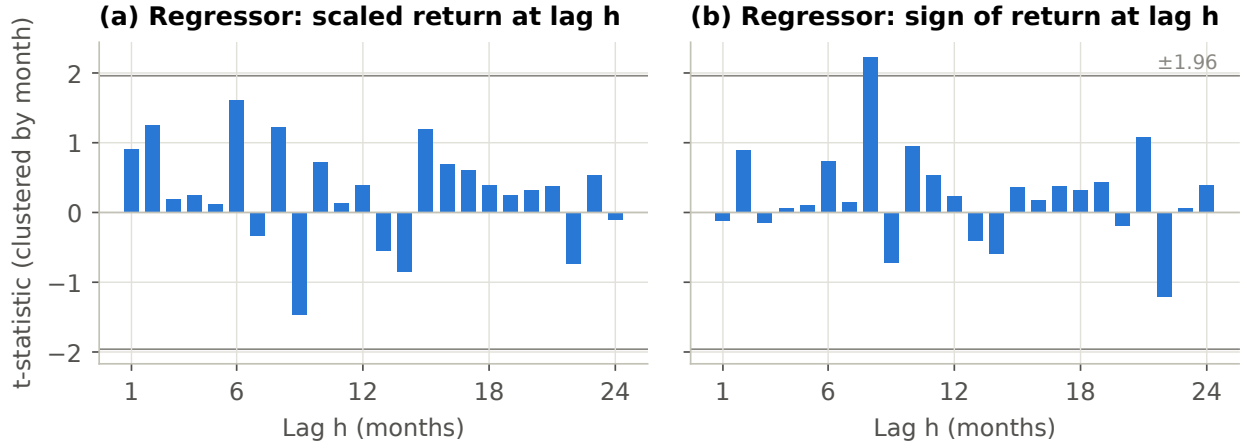


Figure A1: Pooled regressions of the volatility-scaled return on its own value (a) and sign (b) at lag h months. Horizontal lines mark ± 1.96 .

Table B1: Correlations of monthly excess returns, April 2008 to September 2026.

	SPY	EFA	EEM	IEF	TLT	GLD	DBC	UUP	FXY	FXA
SPY	1.00									
EFA	0.87	1.00								
EEM	0.76	0.86	1.00							
IEF	-0.04	0.03	0.00	1.00						
TLT	-0.07	-0.03	-0.04	0.92	1.00					
GLD	0.10	0.21	0.31	0.35	0.27	1.00				
DBC	0.47	0.52	0.53	-0.30	-0.34	0.26	1.00			
UUP	-0.48	-0.70	-0.67	-0.19	-0.10	-0.46	-0.47	1.00		
FXY	-0.04	0.08	0.09	0.53	0.45	0.40	-0.06	-0.42	1.00	
FXA	0.63	0.77	0.82	0.11	0.06	0.40	0.57	-0.78	0.22	1.00

Table B2: Pooled predictive regressions of the volatility-scaled monthly excess return on the lagged k -month signal, 2,220 instrument-months. Standard errors clustered by month.

Signal s_{t-1}^k	Pooled		Instrument fixed effects	
	Coef.	t	Coef.	t
1m	-0.001	-0.12	-0.003	-0.31
3m	0.014	1.36	0.010	0.93
6m	0.013	1.35	0.008	0.78
12m	0.016	1.66	0.010	1.02
Blend	0.019	1.31	0.012	0.74

Table B3: Accuracy of the volatility forecast. Ex-ante vol is the mean forecast at rebalance; the bias statistic is the standard deviation of monthly returns divided by the ex-ante monthly volatility. Given the kurtosis of V2's standardised returns, the standard error of its bias statistic is about 0.05. $|z|$: absolute standardised monthly return.

	Ex-ante vol (%)	Realised vol (%)	Bias statistic	Months $ z > 2$	Months $ z > 3$
V1 raw TSMOM	7.15	7.01	1.00	13	3
V2 vol-targeted	10.00	11.27	1.13	18	3
Long-only risk parity	10.00	11.00	1.10	15	2

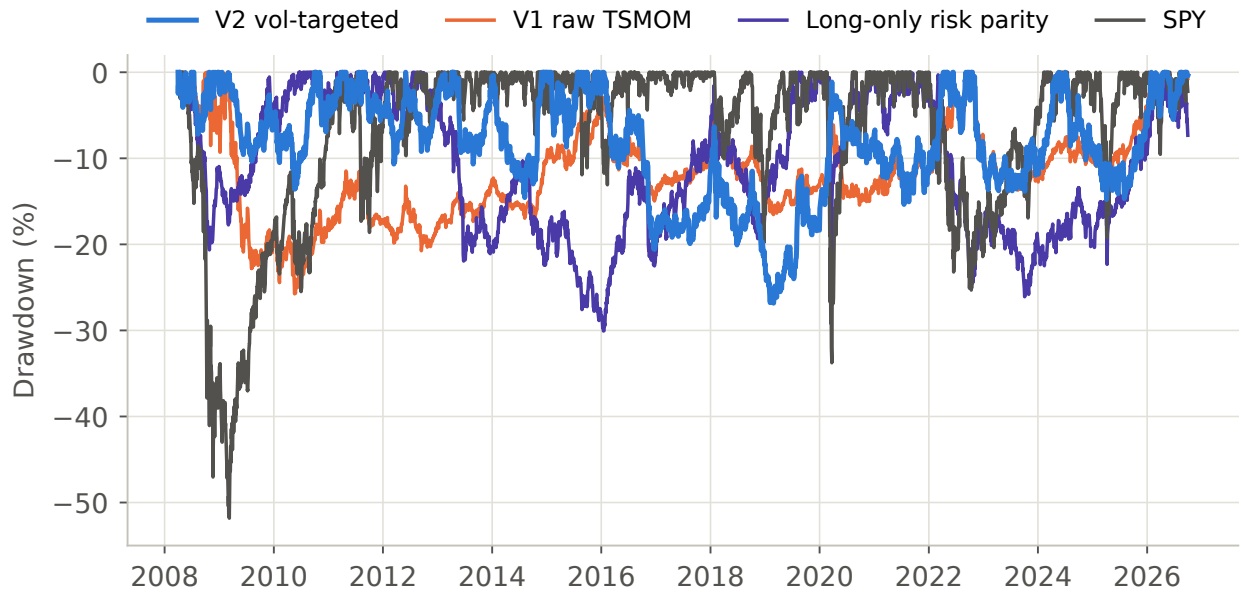


Figure B1: Drawdowns of the cumulative excess return, daily.

Table B4: Average share of ex-ante portfolio variance by asset class (%). V2 per-instrument gives each instrument an equal risk budget instead of each asset class.

Asset class	V1	V2	V2 per-instrument
Equity	48.5	22.8	32.6
Rates	14.0	32.6	22.3
Commodity	22.5	24.0	17.1
FX	15.1	20.6	28.0

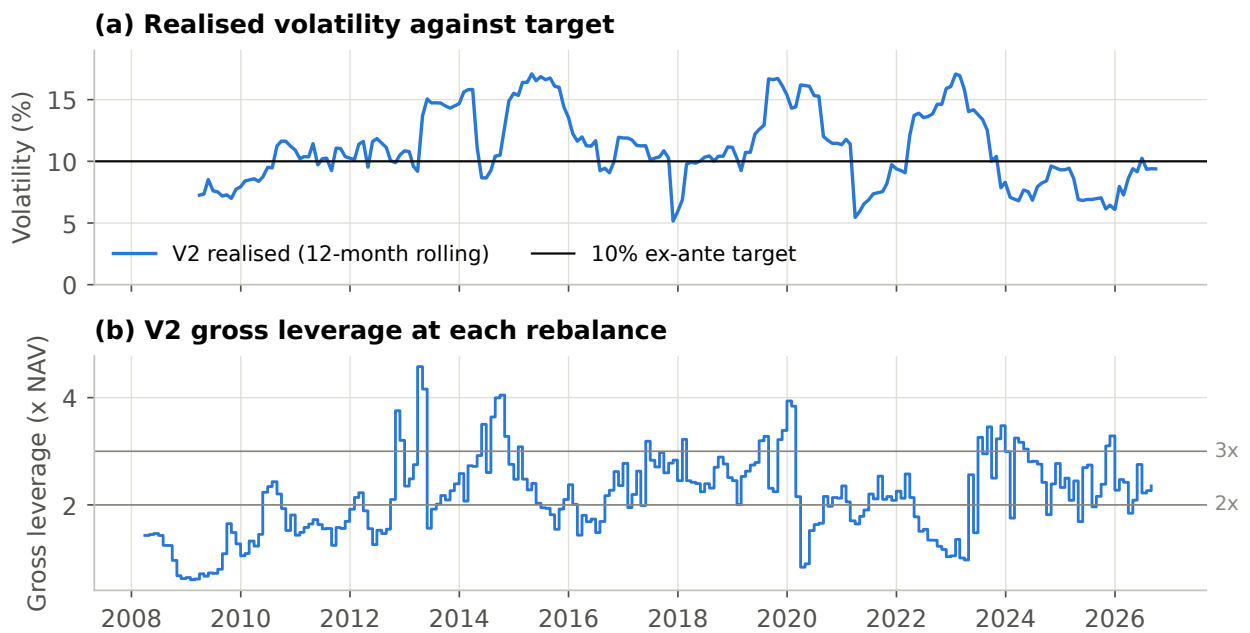


Figure B2: (a) Twelve-month rolling realised volatility of V2 against the 10% ex-ante target. (b) V2 gross leverage at each monthly rebalance.

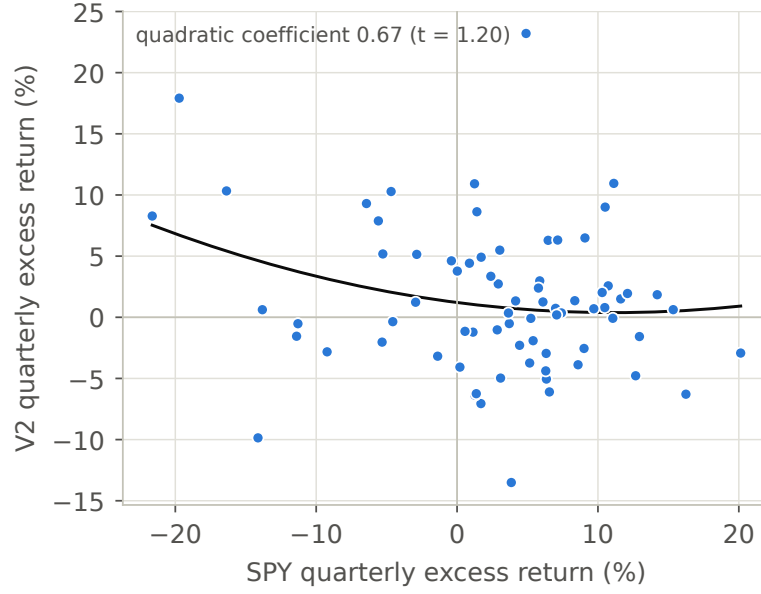


Figure B3: Quarterly V2 excess returns against quarterly SPY excess returns (74 quarters) with the fitted quadratic $r_q = a + b \text{SPY}_q + c \text{SPY}_q^2$; Newey-West t -statistic.

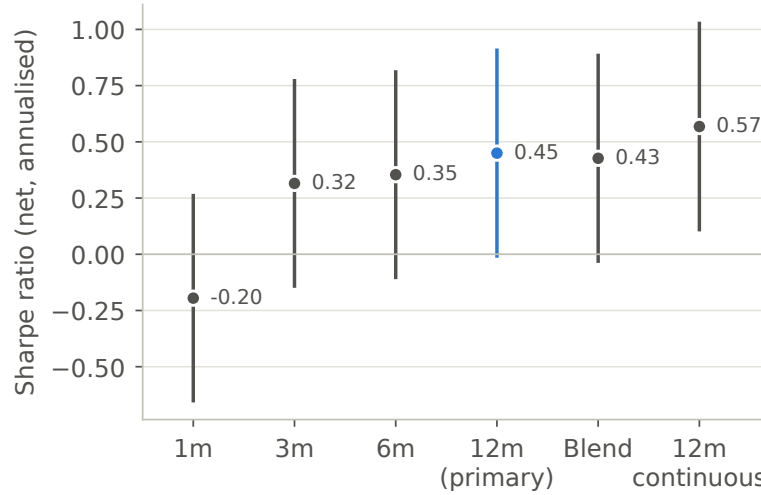


Figure B4: V2 Sharpe ratio by signal, net of costs, with 95% intervals from Lo (2002) standard errors. The intervals are not independent: all variants share the same instruments and sample.

Table B5: Subperiod halves of 111 months each. Annualised mean excess return (%) and Sharpe ratio with Lo (2002) standard error.

	2008-04 to 2017-06			2017-07 to 2026-09		
	Mean	Sharpe	(SE)	Mean	Sharpe	(SE)
V1 raw TSMOM	1.00	0.13	(0.33)	2.58	0.41	(0.33)
V2 vol-targeted	4.36	0.38	(0.33)	5.77	0.52	(0.33)
V3 regime (R1)	3.77	0.27	(0.33)	6.58	0.53	(0.33)
Long-only risk parity	3.44	0.30	(0.33)	2.95	0.28	(0.33)
SPY	9.65	0.63	(0.33)	12.70	0.80	(0.33)
60/40	7.45	0.83	(0.33)	6.86	0.66	(0.33)

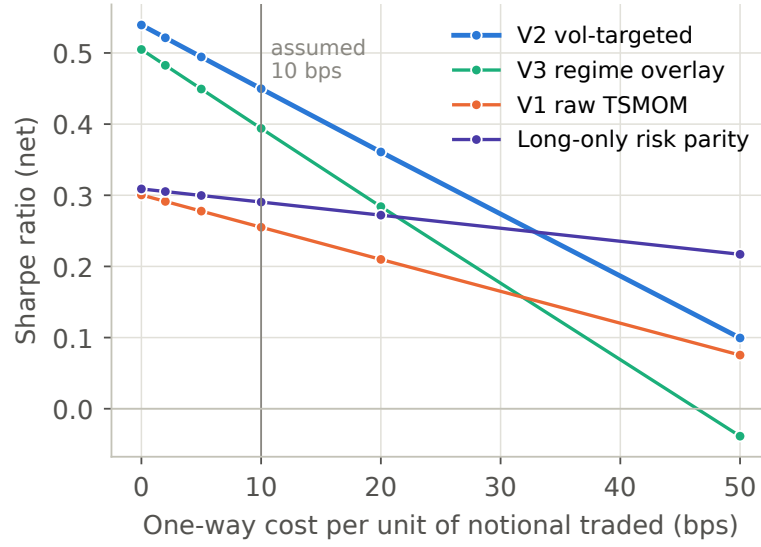


Figure B5: Sharpe ratio against one-way trading cost. The vertical line marks the 10 bps assumption.

Table B6: Cost sensitivity. Annual turnover (multiples of NAV), Sharpe ratio at each one-way cost per unit of notional traded, and the linear break-even cost (annual gross mean excess return divided by annual turnover).

	Turnover	Sharpe ratio at one-way cost (bps)						Break-even (bps)
		0	2	5	10	20	50	
V1 raw TSMOM	3.1	0.30	0.29	0.28	0.26	0.21	0.08	68
V2 vol-targeted	9.8	0.54	0.52	0.49	0.45	0.36	0.10	62
V3 regime (R1)	14.2	0.50	0.48	0.45	0.39	0.28	-0.04	47
Long-only risk parity	1.9	0.31	0.31	0.30	0.29	0.27	0.22	177